

2.1 Evolution

Everything has a “start”. The Internet, as we know it today, also had a very humble but interesting beginning.

J C R Licklider of the Massachusetts Institute of Technology (MIT) envisioned the Internet as far back as August 1962 in a series of memos written by him that talked about social interactions that could be enabled through networking, a concept that he termed his “Galactic Network”.

According to this concept, all computers across the planet would be interconnected and by this, everyone could quickly access data and programs from any ‘site’. This is what actually happens today. J C R Licklider joined DARPA (Defence Advanced Research Projects Agency <http://www.darpa.mil/>) in October 1962 and was its first research head. In due course at DARPA, he convinced his successors, Ivan Sutherland, Bob Taylor, and MIT researcher Lawrence G Roberts, of the importance of this networking concept.

In late 1966, MIT researcher Lawrence G Roberts went to DARPA to develop the computer network concept and quickly put together his plan for the “ARPANET” (Advanced Research Projects Agency Network related to the US Department of Defence) publishing it in 1967. Roberts presented his paper at a conference, where, incidentally, Donald Davies and Roger Scantlebury of NPL (National Physical Laboratory) from the UK presented a paper on a packet network concept.

Earlier during his research, Leonard Kleinrock at MIT convinced Roberts of the feasibility of using packets rather than circuits to transfer data, which, by itself was a major leap forward in the area of computer networking. To prove this, Roberts, with Thomas Merrill in 1965, connected the TX-2 computer in Massachusetts to the Q-32 computer in California using an extremely low-speed dial-up telephone line creating the first (though small) wide-area computer network ever built.